



澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

Faculty of Applied Sciences
Bachelor of Science in Computing / Artificial Intelligence

COMP4299/CSAI4299 Final Year Project Project Proposal

Academic Year 2026/27

An IoT Environmental Monitoring System with Real-Time Alerting
and Spark-Based Distributed Batch Analytics for Server Rooms

Project number: 81
Student ID: P2321251
Student name: Sun WenBin

Supervisor: Tang Su Kit Jacky

Submission Date: Sep 10 , 2026

1 Project Description

Server rooms and data centres house critical IT infrastructure whose reliability depends on strict environmental control. Excessive temperature accelerates hardware wear and can lead to thermal shutdown or permanent component failure, while high humidity causes condensation and corrosion, and very low humidity invites electrostatic discharge. Many small and medium-sized organisations still do not monitor their server environments in real time, relying instead on periodic manual checks or basic thermostat alarms, so a developing fault can remain undetected for hours. This project therefore designs and builds a low-cost, real-time environmental monitoring system that continuously tracks temperature and humidity and notifies operators when conditions leave a safe range.

The cost of unplanned IT downtime is high, and heat- and humidity-related failures are among the most common causes of hardware loss in small server rooms. Real-time monitoring lets operators detect and respond to abnormal conditions before damage occurs, and a complete log of environmental readings supports compliance reporting for standards and audits that require documented environmental control. Commercial monitoring solutions are usually expensive and closed, so a system built from a Raspberry Pi and commodity DHT22 sensors makes continuous monitoring affordable and practical for small organisations.

The project is driven by the following SMART objectives:

Sensor data collection — continuously read temperature (± 0.5 °C accuracy) and humidity (± 2 % RH accuracy) from DHT22 sensors at 1-minute intervals, achieving a ≥ 99 % data capture rate over a 7-day test period.

Web dashboard — build an interactive dashboard with real-time gauges and 24-hour trend charts using Chart.js, with page refresh under 2 seconds.

Alert system — implement configurable threshold alerts (e.g. temperature > 30 °C, humidity > 70 %) that trigger Telegram notifications within 10 seconds of a threshold breach.

Data persistence — store all sensor readings in a SQLite time-series database supporting queries over any 30-day window with a response time under 1 second.

Export and reporting — enable CSV export of historical data for any date range, with report generation completing within 5 seconds for up to 30 days of data.

The deliverable is a working prototype comprising: (i) two or three Raspberry Pi sensor nodes, each equipped with a DHT22 sensor, that capture and transmit readings every minute; (ii) a Flask backend that persists readings in SQLite and exposes REST APIs; (iii) a Chart.js and Bootstrap web dashboard showing live readings and 24-hour trends; (iv) a Telegram alert service that notifies operators within seconds of a threshold breach; and (v) a CSV export facility for compliance reporting. As an extension, a Spark and Hive batch-processing layer aggregates historical readings for longer-term trend analysis.

The main challenges are: (i) DHT22 sensor reliability, since the sensor occasionally returns invalid reads that must be detected and retried; (ii) meeting the $\geq 99\%$ capture target, which requires local buffering and re-transmission when the network link between the Raspberry Pi and the server is temporarily unavailable; (iii) real-time alert latency, because threshold checking must be performed at the point of ingestion for alerts to fire within 10 seconds; (iv) query performance, which depends on indexes and pre-aggregation to keep 30-day queries and CSV exports within the stated time limits; and (v) hardware-dependent issues such as sensor calibration and accuracy verification in a real environment.

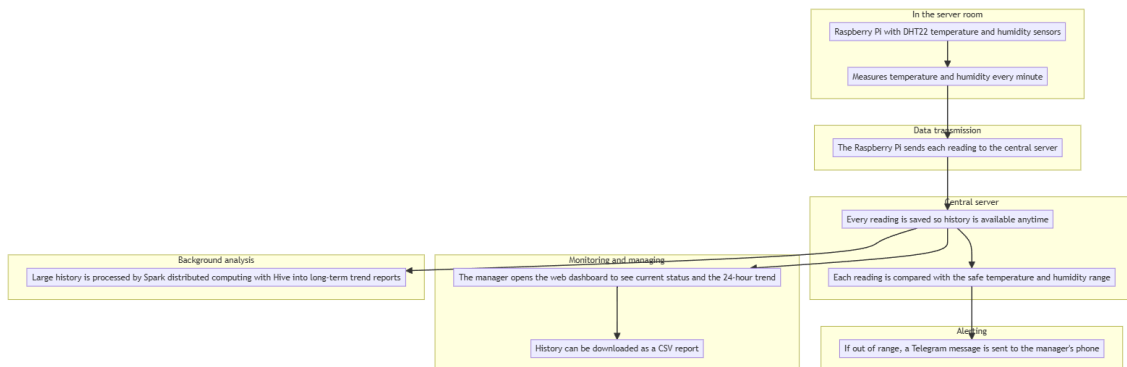


Figure: Logical design of the environmental monitoring system.

2 Summary of Related Work and Key References

Real-time environmental monitoring for computing facilities is well established. ASHRAE TC 9.9 publishes widely used thermal guidelines that define recommended temperature and humidity envelopes for data processing environments, and these safe ranges directly inform the alert thresholds configured in this project [4]. Commercial server-room monitoring suites offer accurate multi-point sensing but are typically expensive and closed, which places them out of reach of many small and medium-sized organisations.

On the low-cost side, the Internet of Things has made embedded sensing widely accessible: Gubbi et al. describe the vision and architectural elements of IoT systems in which low-power devices such as the Raspberry Pi collect and forward physical measurements [3]. Many open hardware projects combine a Raspberry Pi with a DHT22 sensor to log temperature and humidity, using the sensor datasheet as the reference for accuracy and timing requirements [1][2].

On the software side, lightweight REST back-ends built with Flask [7] and embedded relational storage such as SQLite [8] are common choices for small telemetry services because they add little overhead while providing fast, indexed queries. Interactive dashboards are typically rendered in the browser with Chart.js [5], and instant notifications can be delivered through the Telegram Bot API at low cost and with minimal setup [6]. For longer-term analysis, batch processing frameworks such as Apache Spark and the Hive warehouse [9][10] can aggregate large histories of readings efficiently.

This project combines these established elements into a single low-cost system: commodity DHT22 sensors on Raspberry Pi nodes, a Flask and SQLite backend, a Chart.js dashboard, Telegram alerts and, as an extension, a Spark/Hive batch-analysis layer. Its contribution is the integration and validation of these components against measurable targets — capture rate, alert latency, and query and export time — in a server-room scenario.

3 Preliminary Schedule

The schedule below follows the official FYP calendar. All major deliverables are aligned to the stated submission deadlines. Activities A9 and A10 run in parallel and are completed before Week 6 of the second semester (25 Feb). The Final Presentation and Poster Presentation take place in Week 10 (Mar). The Final Report, demo video and optional conference-ready manuscript are submitted in Week 13 (15 Apr).

Activity	Description	Duration	Predecessor
A1	Literature review and requirements analysis	3 wks (W1-3)	-

A2	Project planning; write Chapter 3 of Progress Report	1 wk (W5)	A1
A3	Proposal and ethics form submission	2 wks (by 10 Sep)	A1
A4	Hardware setup: Raspberry Pi + DHT22, sensor script and calibration	4 wks (W6-9)	A1, A3
A5	Backend development: Flask ingestion API, SQLite persistence, threshold checking, Telegram alerts	4 wks (W9-12)	A4
A6	Progress Report submission and presentation	2 wks (W12-13)	A2, A5
A7	Dashboard development: Chart.js real-time gauges, 24-hour trends, CSV export	2 wks (W14-15)	A5
A8	Address supervisor/assessor comments on Progress Report	1 wk (W15)	A6
A9	System integration, 7-day continuous test, SMART-objective validation	6 wks (S2 W1-6)	A7
A10	Spark/Hive batch layer (extension): Hive tables, PySpark aggregation job	3 wks (S2 W3-5)	A7, A8
A11	Final Report draft submission	2 wks (S2 W8)	A9, A10
A12	Poster submission; Final Presentation and Poster Presentation	2 wks (S2 W9-10)	A11
A13	Revise Final Report based on presentation feedback; prepare demo video	2 wks (S2 W11-12)	A12

A14	Final Report submission, demo video submission, and optional conference-ready manuscript	1 wk (S2 W13)	A13
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Key deliverables:

D1 - Project Proposal & Ethics Checklist - 10 Sep 2026

D2 - Progress Report & Progress Presentation - 12 / 19 Nov 2026

D3 - Final Presentation & Poster Presentation - Week 10 (Mar 2027)

D4 - Final Report, Demo Video & (optional) Conference Manuscript - 15 Apr 2027

4 Risk Assessment

Risk 1: Invalid or inaccurate DHT22 readings

- **Description:** The DHT22 occasionally returns out-of-range or spurious values. Such readings would corrupt the dataset and could break the accuracy (± 0.5 °C, ± 2 % RH) and capture-rate (≥ 99 %) objectives.
- **Probability of occurrence:** medium
- **Impact:** Medium. It corrupts data quality and threatens the verification targets.
- **Response:** Validate and retry each read, reject out-of-range values in the collection script, cross-check readings across multiple sensors, and calibrate against a reference thermometer and hygrometer.

Risk 2: Unstable network link between the Raspberry Pi and the server

- **Description:** Readings are transmitted from the Raspberry Pi to the central server over the local network. Intermittent connectivity drops readings and directly threatens the ≥ 99 % capture-rate target.
- **Probability of occurrence:** medium
- **Impact:** High. A single network outage window directly violates the primary capture-rate objective.
- **Response:** Buffer readings locally on the Raspberry Pi and re-transmit them when connectivity recovers; keep a local CSV copy as a fallback.

Risk 3: Alert delivery delay or failure

- **Description:** Threshold alerts are delivered through the Telegram Bot API. API unavailability or network delays could prevent the 10-second alert target from being met.
- **Probability of occurrence:** low
- **Impact:** Medium. Delayed alerts reduce the usefulness of the monitoring system but do not affect data collection.
- **Response:** Evaluate thresholds at the point of ingestion, retry sending with backoff, and use SMTP email as a secondary notification channel.

Risk 4: Server downtime or SQLite data corruption

- **Description:** The server runs on a single virtual machine. A crash or database corruption would lose the stored history and break the query and export functions.
- **Probability of occurrence:** low
- **Impact:** High. Losing all stored readings undermines the data-persistence and reporting objectives.
- **Response:** Enable SQLite WAL mode, schedule regular database backups, and retain the appended CSV as a secondary record.

Risk 5: Query or export performance targets missed

- **Description:** Large date-range queries on the readings table could exceed the 1-second query and 5-second export limits.
- **Probability of occurrence:** low
- **Impact:** Medium. It fails the performance objectives but does not stop the system from functioning.
- **Response:** Index the timestamp column, pre-aggregate data, and paginate large exports.

Priority	Risk Identifier and Description
1	Risk 2: Unstable network link between the Raspberry Pi and the server

2	Risk 4: Server downtime or SQLite data corruption
3	Risk 1: Invalid or inaccurate DHT22 readings
4	Risk 3: Alert delivery delay or failure
5	Risk 5: Query or export performance targets missed

Probability	High			
	Medium			
	Low	Risk3 Risk4 Risk5	Risk1 Risk2	
		Low	Medium	High
		Impact		

References

[1] Aosong Electronics. AM2302 / DHT22 Digital Temperature and Humidity Sensor Datasheet. Aosong Electronics Co., Ltd., 2012.

[2] Raspberry Pi Foundation. Raspberry Pi Documentation.
<https://www.raspberrypi.com/documentation/>, 2026.

[3] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami. Internet of Things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems*, 29(7):1645–1660, 2013.

[4] ASHRAE TC 9.9. 2011 Thermal Guidelines for Data Processing Environments – Expanded Data Center Classes and Usage Guidance. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, 2011.

[5] Chart.js. Chart.js Documentation. <https://www.chartjs.org/docs/>, 2026.

[6] Telegram Messenger LLP. Bot API. <https://core.telegram.org/bots/api>, 2026.

[7] Pallets. Flask Documentation. <https://flask.palletsprojects.com/>, 2026.

[8] SQLite. SQLite Documentation. <https://www.sqlite.org/docs.html>, 2026.

[9] Apache Software Foundation. Apache Spark Documentation. <https://spark.apache.org/docs/latest/>, 2026.

[10] Apache Software Foundation. Apache Hive Documentation. <https://hive.apache.org/>, 2026.

Ethics Checklist

Name of student: <i>(leader of the team, if applicable)</i>	Sun WenBin
Student number:	P2321251
Programme:	☒ BSc in Computing ☐ BSc in AI ☐ MSc in Big Data and IoT ☐ PhD in Computer Applied Technology
Module:	Final Year Project
Title of project:	A Low-Cost IoT Environmental Monitoring System for Server Rooms Using Raspberry Pi and DHT22 Sensors
Supervisor:	Tang Su Kit Jacky

Instructions

Answer the following questions honestly. If your answer falls into a shaded box, you must provide additional information on how you will handle the related ethics issues on the last page of this form.

Project summary

This project designs and builds a low-cost, real-time environmental monitoring system for server rooms. Raspberry Pi sensor nodes equipped with DHT22 sensors continuously measure temperature (± 0.5 °C accuracy) and humidity (± 2 % RH accuracy) at 1-minute intervals and transmit the readings to a central Flask server, where the data is stored in a SQLite database. An interactive web dashboard built with Chart.js and Bootstrap displays real-time gauges and 24-hour trend charts. Configurable threshold alerts send instant Telegram notifications when temperature or humidity exceeds safe limits, and historical data can be exported as CSV for compliance reporting. As an extension, a Spark and Hive batch-processing layer aggregates readings for longer-term trend analysis. The project targets a ≥ 99 % data capture rate over a 7-day test period, page refresh under 2 seconds, alerts within 10 seconds of a threshold breach, and 30-day queries in under 1 second.

Participants (Consent procedures and possible harm)

		Yes	No	N/A
1	Does your project include human participants? If no, you can skip the rest of this section.		✓	
2	Will any of the participants be from vulnerable groups, e.g. children under 18, people with communication difficulties, patients in hospital? If yes, please specify: _____			✓
3	Will you tell participants that their participation is voluntary?			✓
4	Will you obtain written consent for participation?			✓
5	Will you tell participants that they may withdraw from the research at any time and for any reasons without penalty?			✓
6	Is there any realistic risk that any participants could experience either physical or psychological distress or discomfort?			✓

Data Protection

		Yes	No	N/A
7	Will any non-anonymised and/or personalized data be generated and/or stored?		✓	
8	Will you have access to documents containing sensitive data about living individuals?		✓	

Researcher Safety

		Yes	No	N/A
9	Will you be exposed to any risks greater than those encountered in their normal study/working life? E.g. operation of non-standard equipment or extended		✓	
10	Will you be exposed to highly addictive or illegal activities (e.g. gambling, drugs) during the research?		✓	

No ethical issues are identified. The project monitors only physical environmental variables (temperature and humidity) in a server room. It involves no human participants, no personal or anonym sable data, and no activities that pose risks beyond ordinary study or working life.

Any other issues?

Declaration

Student

I have read the instruction carefully and reported honestly the potential ethics issues about the project.

San Wen Bin

Date: Sep 10, 2026

Supervisor

To the best of my knowledge, the student has reported honestly the potential ethics issues about his/her project.

Date: _____

This form is largely based on the ethical approval documents from Cardiff University <https://www.cs.cf.ac.uk/ethics/> and University of Glasgow <http://www.dcs.gla.ac.uk/ethics/>